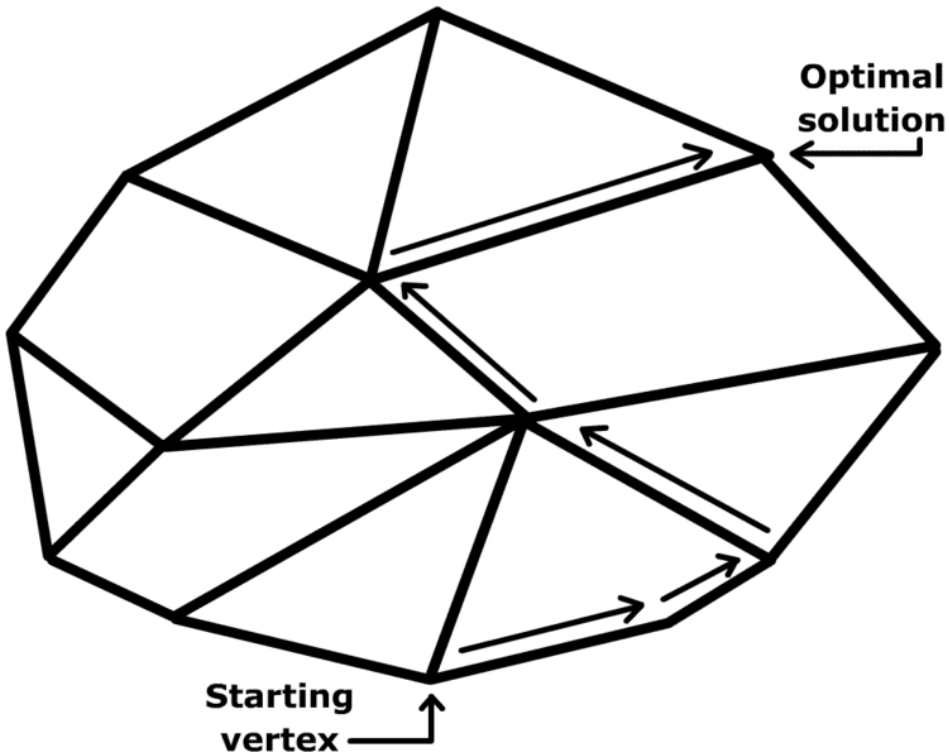
Algorithms and Analysis

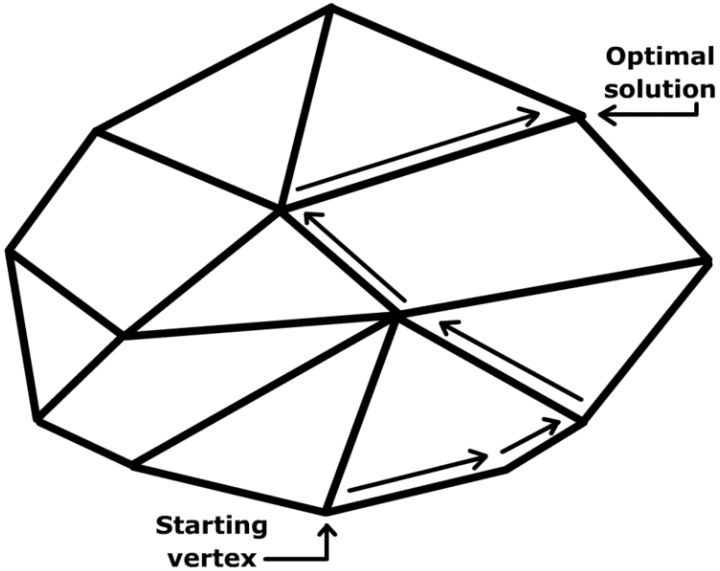
Lesson 28: *Solving Linear Programs*



linear programming, simplex methods, iterative search

Outline

1. **Recap**
2. Basic Feasible Solutions
3. Simplex Method
4. Classic LP Problems



Recap

- Linear programs are problems that can be formulated as follows

$$\min_{\boldsymbol{x}} \boldsymbol{c} \cdot \boldsymbol{x}$$

subject to

$$\mathbf{A}^{\leq}\boldsymbol{x} \leq \boldsymbol{b}^{\leq}, \quad \mathbf{A}^{\geq}\boldsymbol{x} \geq \boldsymbol{b}^{\geq}, \quad \mathbf{A}^{=}\boldsymbol{x} = \boldsymbol{b}^{=}, \quad \boldsymbol{x} \geq \mathbf{0}$$

- Where $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$

- $\mathbf{A}^*$ are matrices and we interpret the inequalities to mean

$$\forall k \qquad \sum_{j=1}^{n} A^{\leq}_{kj} x_j \leq b^{\leq}_k$$

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Optima and Vertices

- Because the objective function is linear $(\boldsymbol{c} \cdot \boldsymbol{x})$ there is a direction where the objective is always improving
- Thus, the optima cannot lie in the interior of the search space
- When we meet a constraint that limits the direction we can move, but we can still move along the constraint
- We then meet another constraint which restricts the direction we can move by two degrees of freedom
- Eventually, we will reach n constraints which defines a vertex of the feasible region and is optimal

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Transforming Linear Programs

- We can always transform an inequality constraint into an equality constraint by adding slack variables

- E.g.

$$\begin{array}{lcll} \boldsymbol{a}_1 \cdot \boldsymbol{x} \geq 0 & \Rightarrow & \boldsymbol{a}_1 \cdot \boldsymbol{x} - z_1 = 0 & z_1 \geq 0 \\ \boldsymbol{a}_2 \cdot \boldsymbol{x} \leq 0 & \Rightarrow & \boldsymbol{a}_2 \cdot \boldsymbol{x} + z_2 = 0 & z_2 \geq 0 \end{array}$$

  z_1 (the excess) and z_2 (the deficit) are known as slack variables

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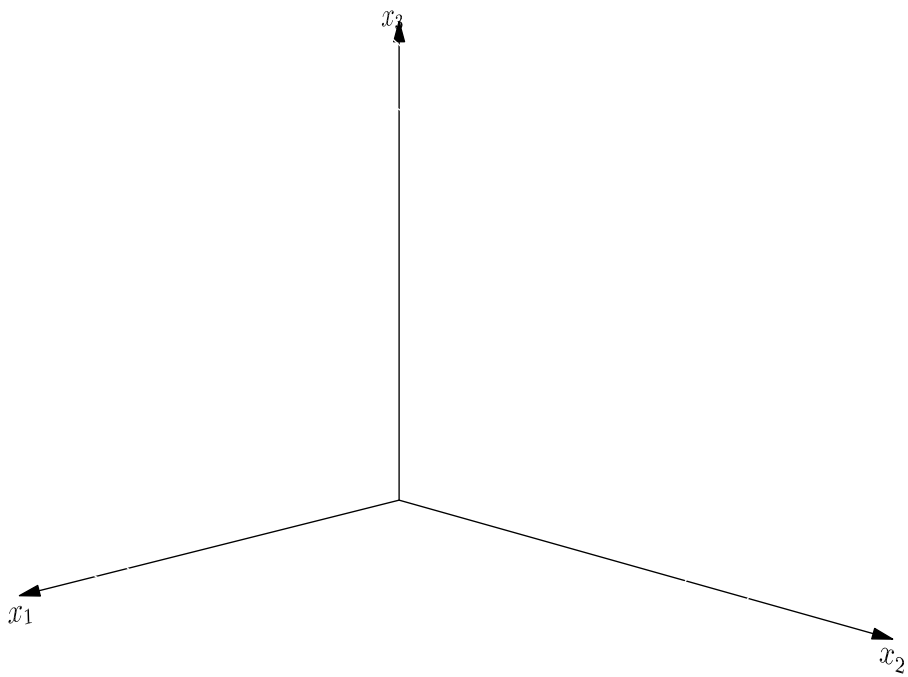

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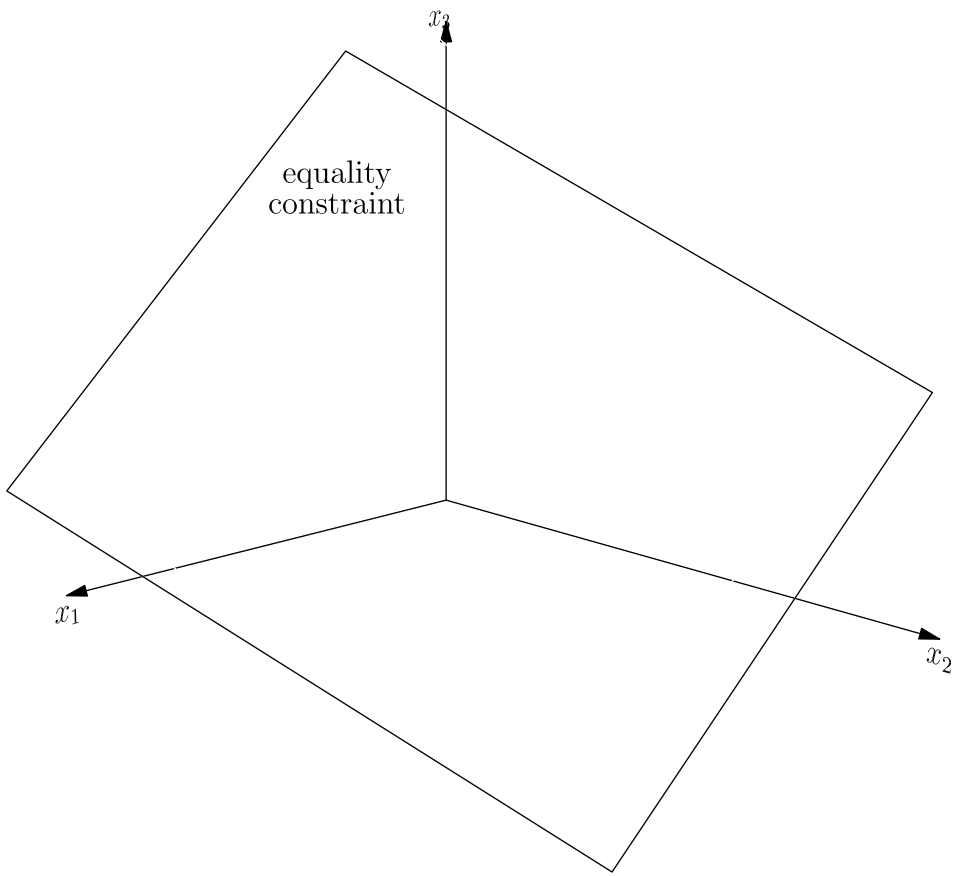

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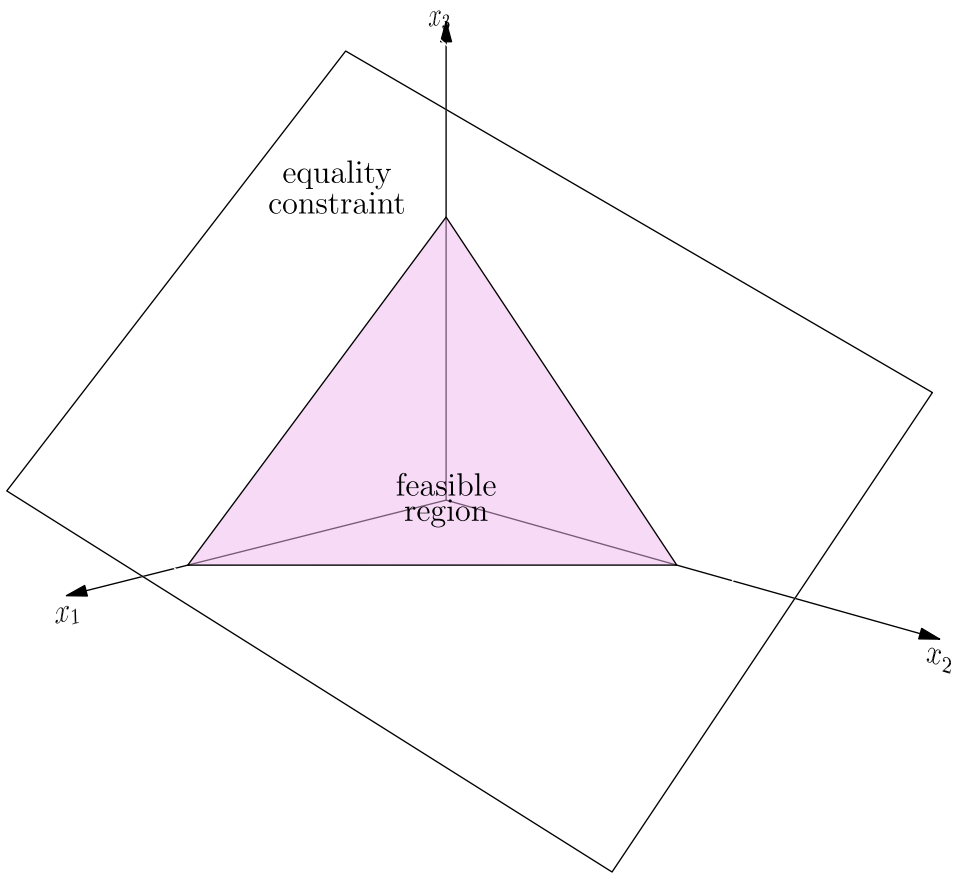

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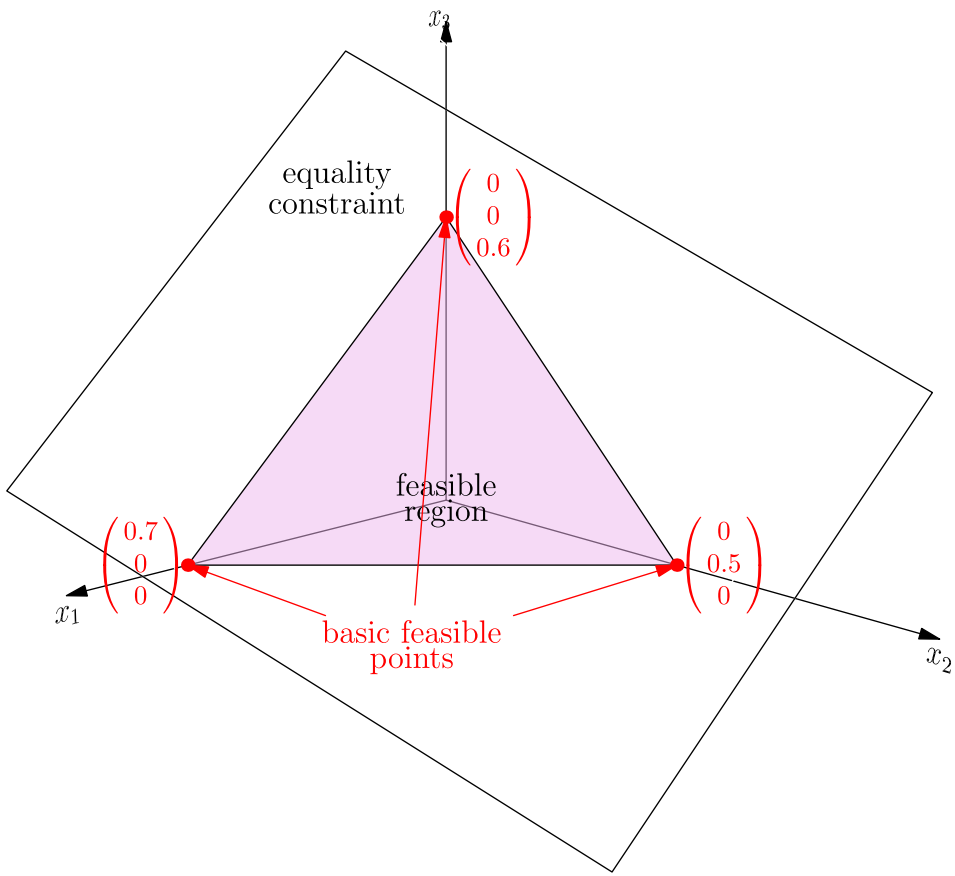

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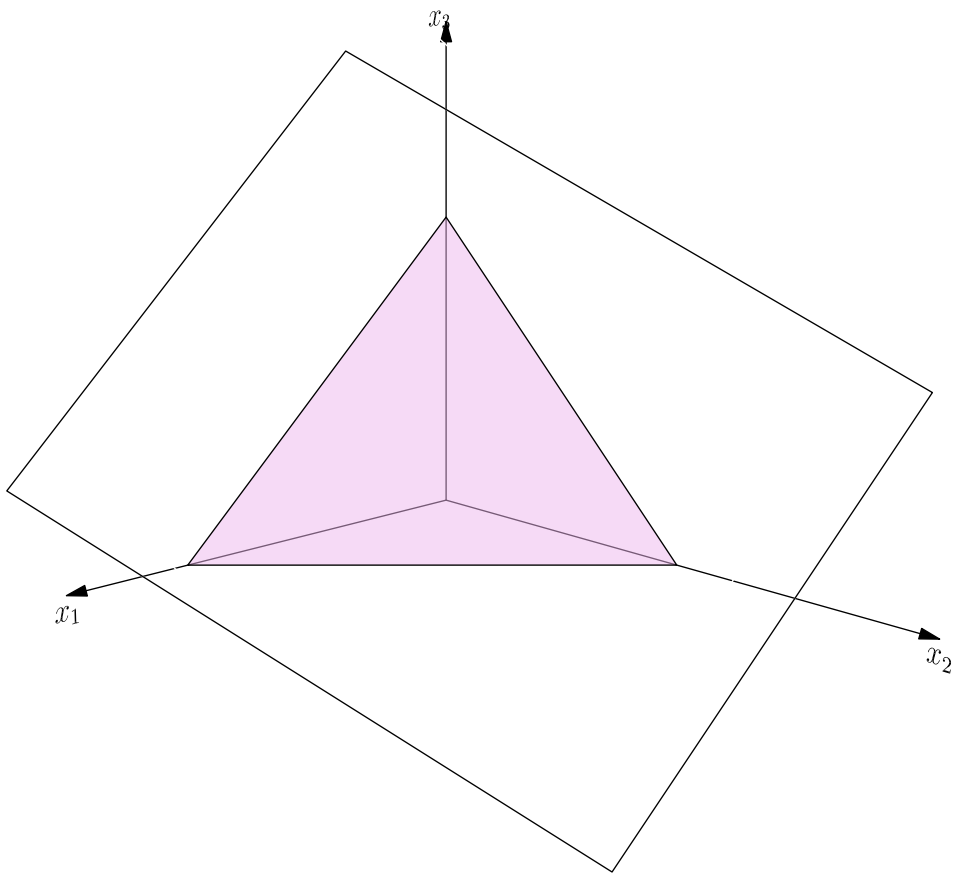

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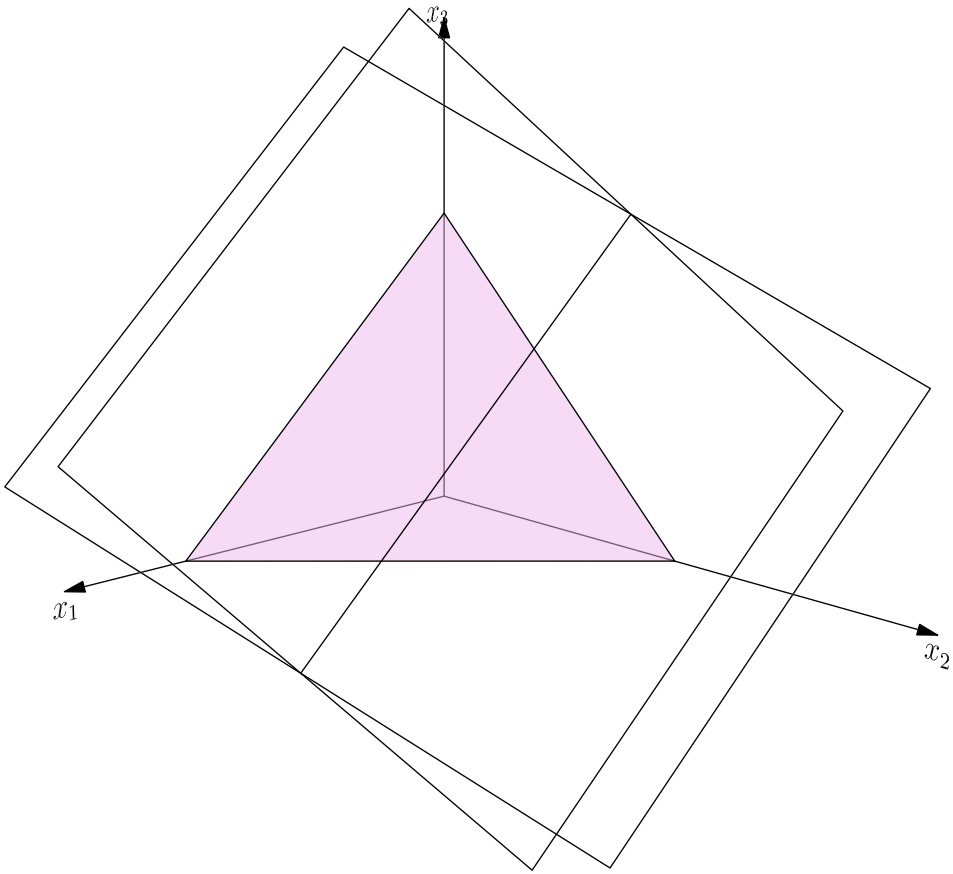

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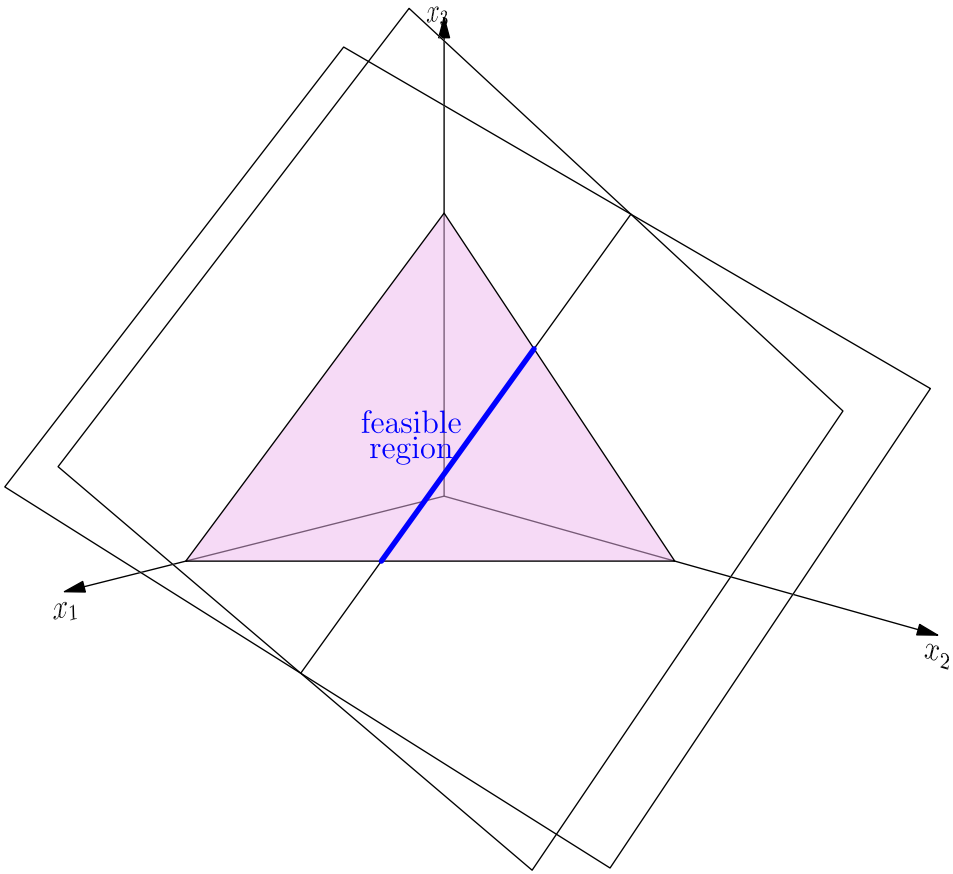

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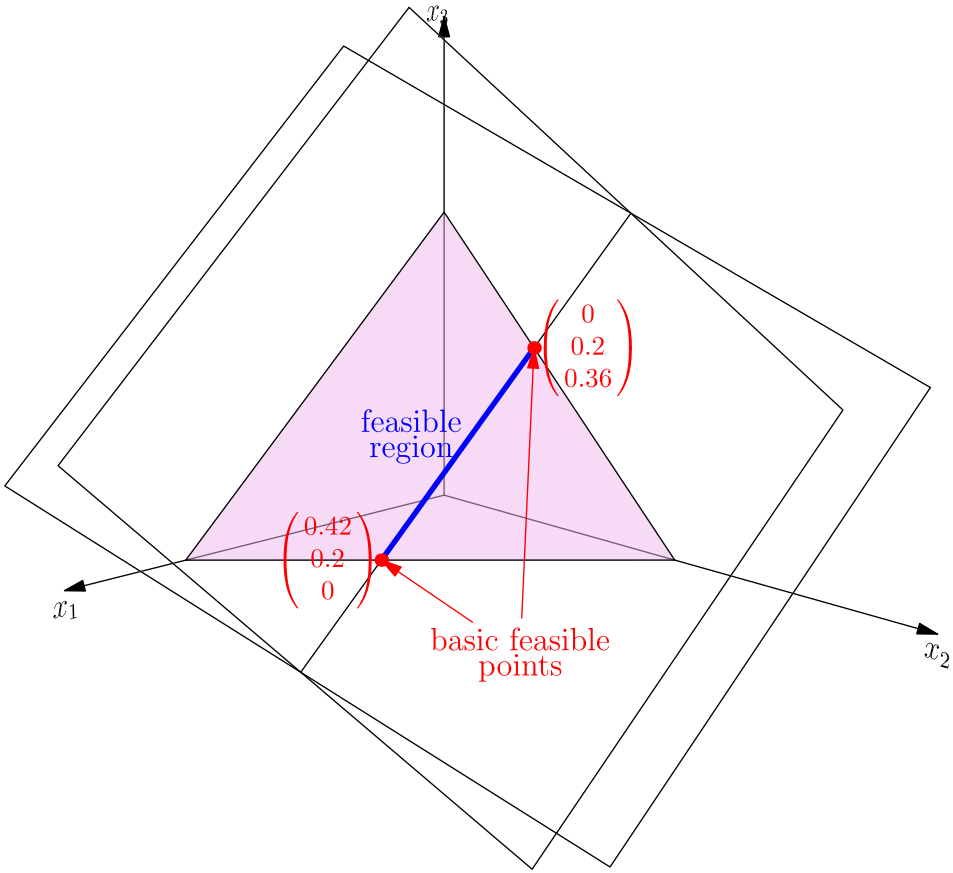

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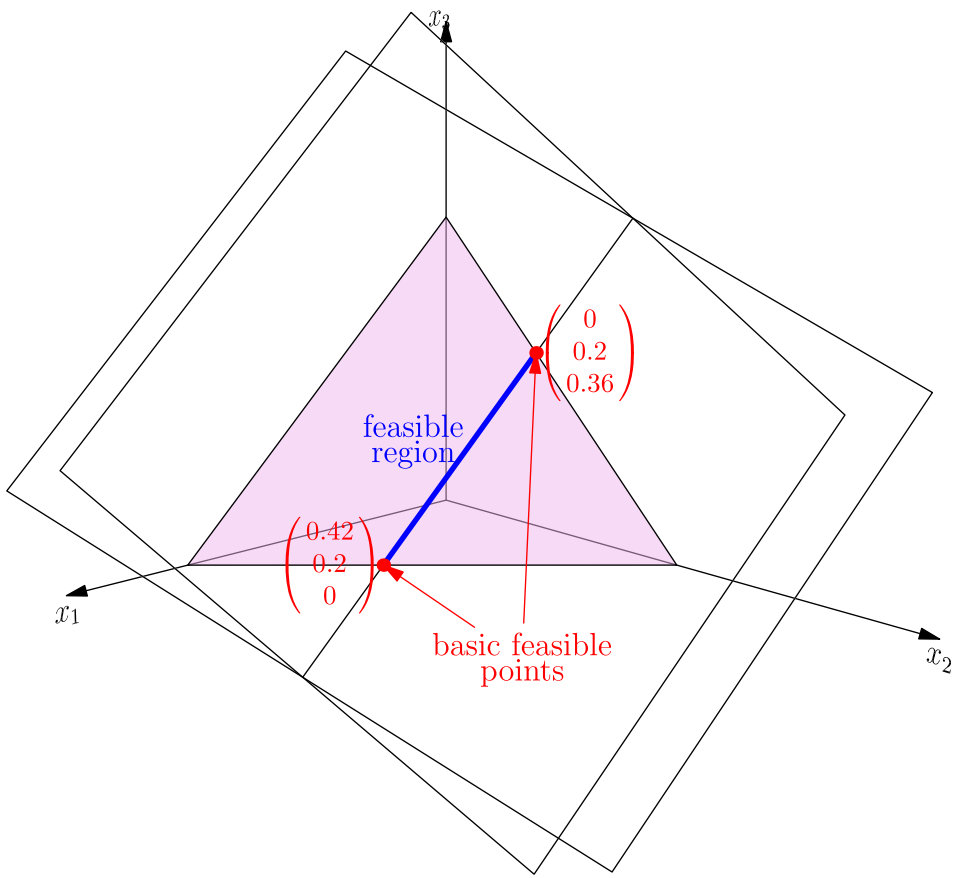

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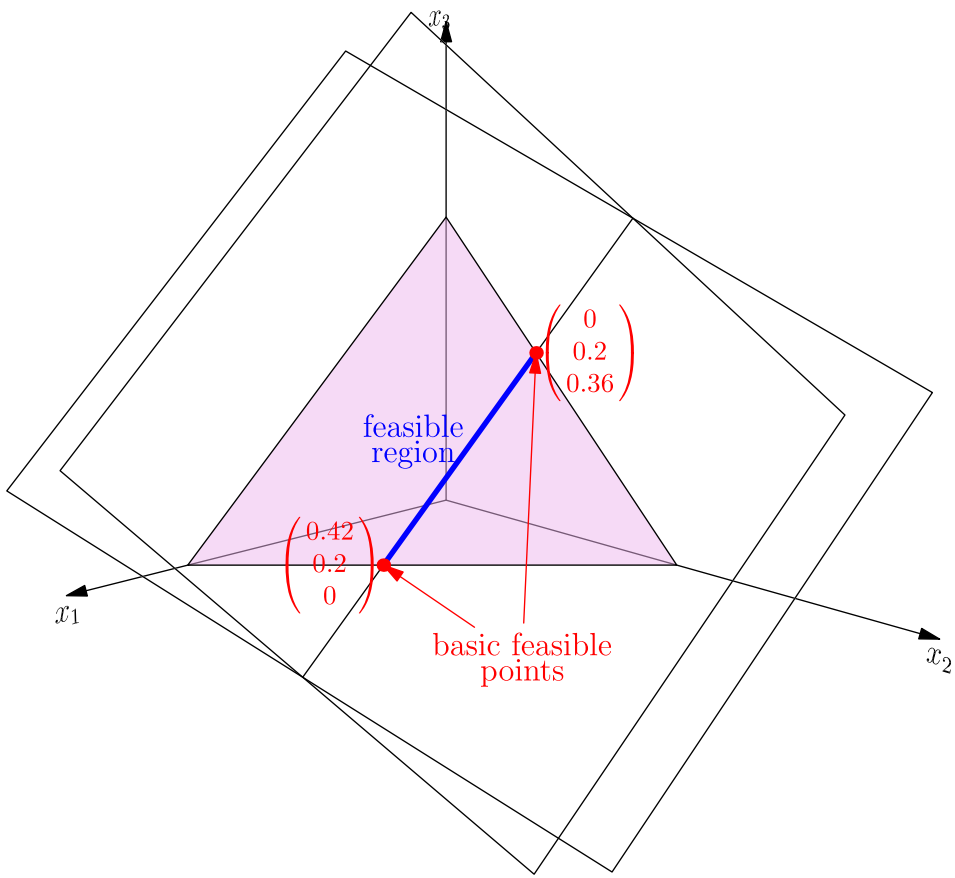

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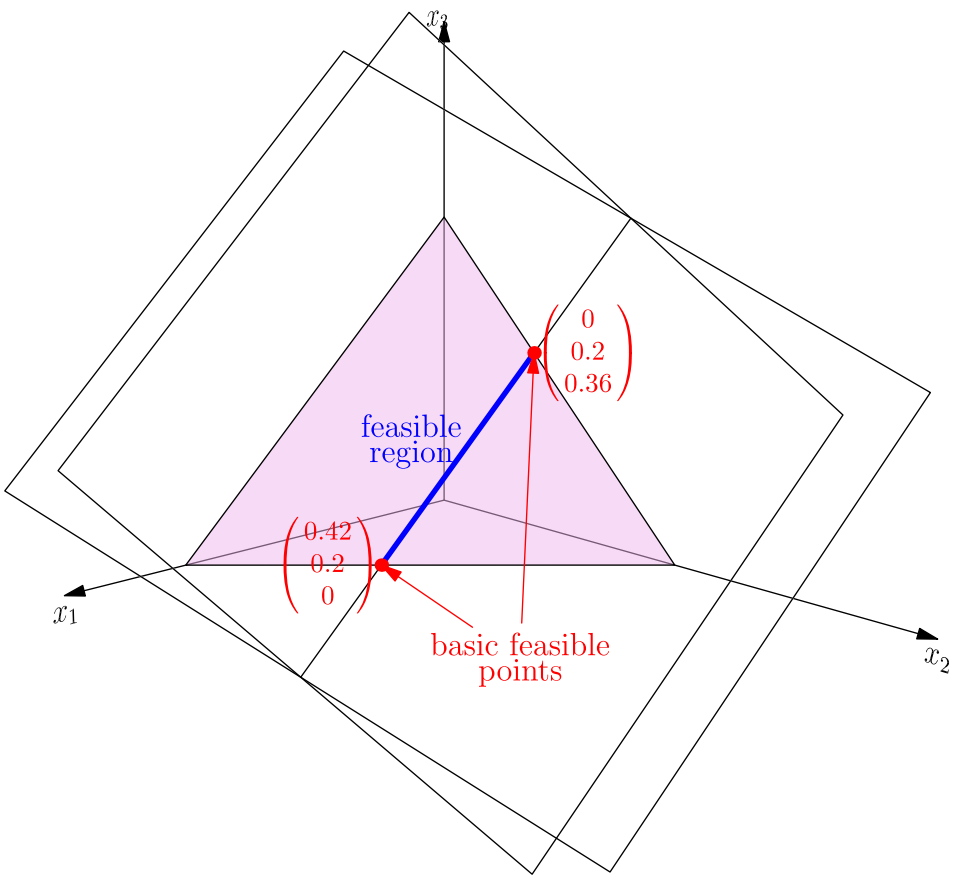

- The basic feasible points for LP problems with n variables and m constraints have at least $n - m$ zero variables
- Typical number of basic feasible solutions is $\binom{n}{m} \geq \left(\frac{n}{m}\right)^m$
- Simplex algorithm organises iterative search for global solutions

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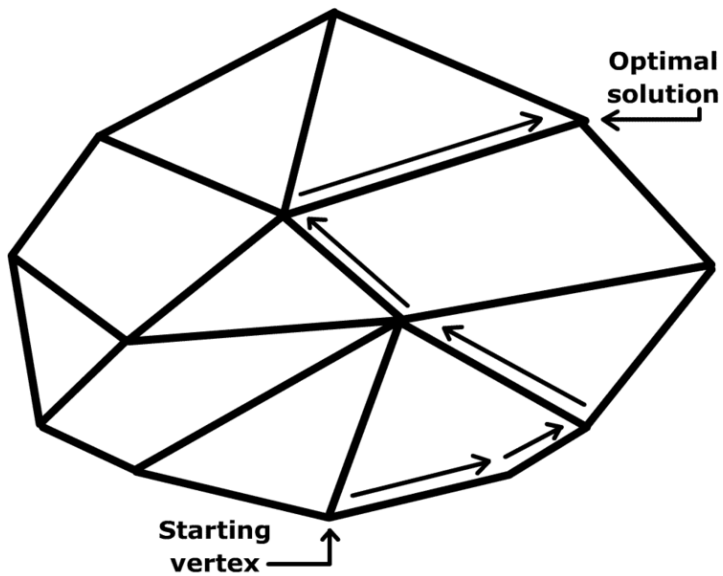

Basic Feasible Solution

- A *basic feasible solution* or *basic feasible point* is a solution that lies at a vertex of the feasible space
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Constraints

- There are two types of constraints

  1. n non-negativity constraints $x_i \geq 0$
  2. m additional constraints, which we can take to be equalities $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$

- Note that some of the variables might be slack variables

- We consider the case when there are more variables than additional constraints, i.e. $n > m$

- This is usually be the case, but. . .

- If this isn't true it turns out you can consider an equivalent problem (dual problem) where you have a variable for each constraint and a constraint for each variable

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Basic Variable

- In total we have n equality and m non-negativity constraints
- n constraints must be satisfied to be at a vertex of feasible region
- So at least $n - m$ of the non-negativity constraints are satisfied (i.e. $x_i = 0$)
- The $n - m$ variables that are zero are said to be **non-basic variables**
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Initial Basic Feasible Solution

- One of the tricky bits of tackling a linear program is to find an initial feasible solution

- We do this in **phase one** of the simplex program

- To do this for each additional constraint we add a new **auxiliary variable** ξ_k, e.g.

$$\forall\, k \in \{1, 2, \ldots, m\} \qquad \xi_k + \sum_i A_{ki} x_i = b_k \geq 0$$

- We then can find a basic feasible solution by setting $x_i = 0$ so

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Eliminating Auxiliary Variables

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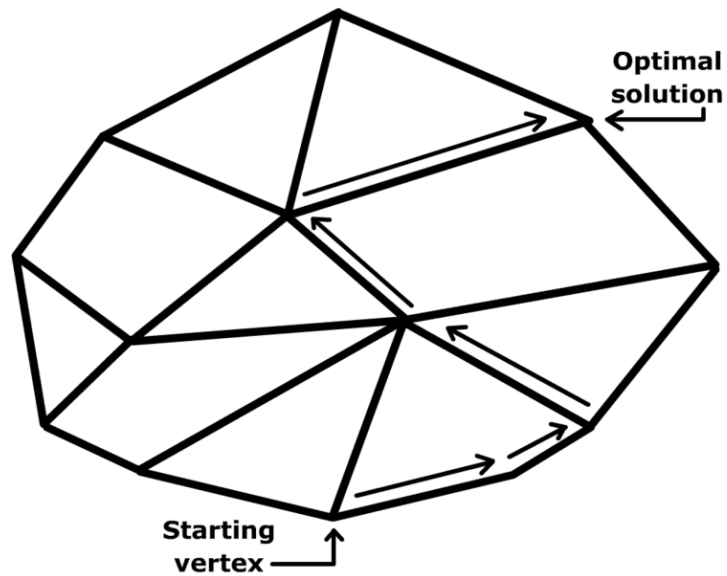

Phase Two

- In phase two we now have an initial basic feasible solution (with $n - m$ zero variables)
- We then run the simplex algorithm on the original objective function $f(\boldsymbol{x}) = \boldsymbol{c} \cdot \boldsymbol{x}$
- That is we move to a neighbouring vertex which gives the best increase in the objective function
- To help organise this search we write the objective function and constraints in a **restricted normal form** and then build a **tableau** showing the *basic variables* and the *non-basic variables*

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Restricted Normal Form

- To perform the moves between vertices it helps to represent the problem in a **restricted normal form**
- Starting from a basic feasible point we have a constraint for each basic (non-zero) variable
- We write the constraints as an equality between basic and non-basic (zero valued) variables
- Similarly we write the objective function in terms of non-basic variables
- This is always possible as we can use the constraints to eliminate the basic variables

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Tableau

$$\max_{\mathbf{x}} \; f(\mathbf{x}) = -8 + 5x_1 + 7x_4 + 3x_5 + 8x_7 - 6x_8 - 3x_9$$

where

$$x_2 = 3 + 6x_1 + 6x_4 - 5x_5 - 5x_7 + 5x_8 - 5x_9 \geq 0$$

$$x_3 = 1 - 5x_1 - 8x_4 - 8x_5 + 4x_7 - 8x_8 - 5x_9 \geq 0$$

$$x_6 = 8 - 8x_1 - 7x_4 - 2x_5 - 5x_7 - 3x_8 + 4x_9 \geq 0$$

$$x_1 = x_4 = x_5 = x_7 = x_8 = x_9 = 0$$

Tableau

$$\max_{\mathbf{x}} \; f(\mathbf{x}) = -8 + 5x_1 + 7x_4 + 3x_5 + 8x_7 - 6x_8 - 3x_9$$

where

$$x_2 = 3 + 6x_1 + 6x_4 - 5x_5 - 5x_7 + 5x_8 - 5x_9 \geq 0$$

$$x_3 = 1 - 5x_1 - 8x_4 - 8x_5 + 4x_7 - 8x_8 - 5x_9 \geq 0$$

$$x_6 = 8 - 8x_1 - 7x_4 - 2x_5 - 5x_7 - 3x_8 + 4x_9 \geq 0$$

$$x_1 = x_4 = x_5 = x_7 = x_8 = x_9 = 0$$

		x_1	x_4	x_5	x_7	x_8	x_9
$f(\mathbf{x})$	-8	5	7	3	8	-6	-3
x_2	3	6	6	-5	-5	5	-5
x_3	1	-5	-8	-8	4	-8	-5
x_6	8	-8	-7	-2	-5	-3	4

Awkward Problems

- If there are any column with all entries positive then this variable can be increase forever—this is a signal that the linear programming problem is unbounded
- You can also find that a basic variable becomes zero—this is known as a degenerate feasible vector
- It can by removed by exchanging variables on the left of the inequality with variables on the right
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High Performance Solvers

- Although the tableau method is the "classic solver" it doesn't cut the mustard for large scale problems
- The simplex update can also be viewed as solving a linear set of equations which is facilitated by performing an LU-decomposition
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Time Complexity of Simplex

- The time complexity of the updates is $O(n^2)$
- The critical question is how many updates are necessary
- It turns out that typically this is $O(n)$ making the simplex algorithm $O(n^3)$
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Interior Point Method

- An alternative to the simplex method is the interior point method which always remains in the feasible region, away from the constraints
- These method iterate towards the constraints and are provably polynomial
- For small linear programming problems they are out-performed in practice by the simplex method
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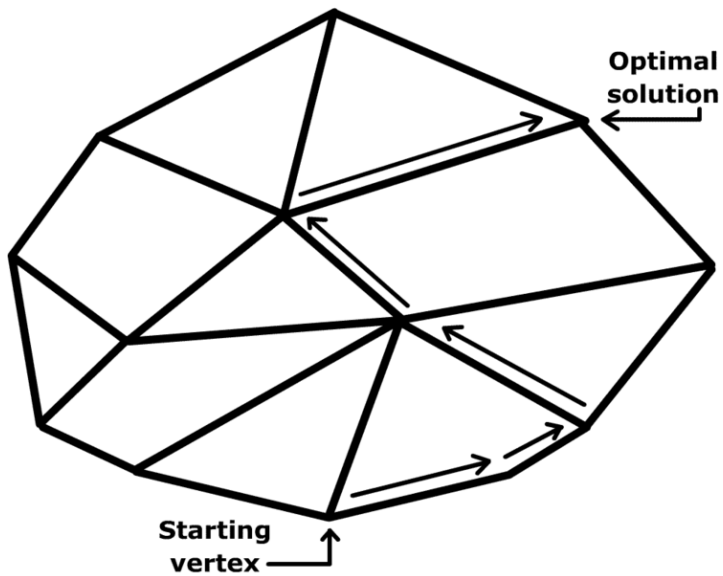
Outline

1. Recap
2. Basic Feasible Solutions
3. Simplex Method
4. **Classic LP Problems**



LP Problems

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- One way is just to feed it to a LP-solver
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- We consider a couple of classic problems: *maximum flow* and *linear assignment*

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Maximum Flow

- In maximum flow we consider a directed graph representing a network of pipes
- We choose one vertex as the source and a second vertex as a sink
- Each edge has a flow capacity that cannot be exceeded
- The problem is to maximise the flow between source of sink
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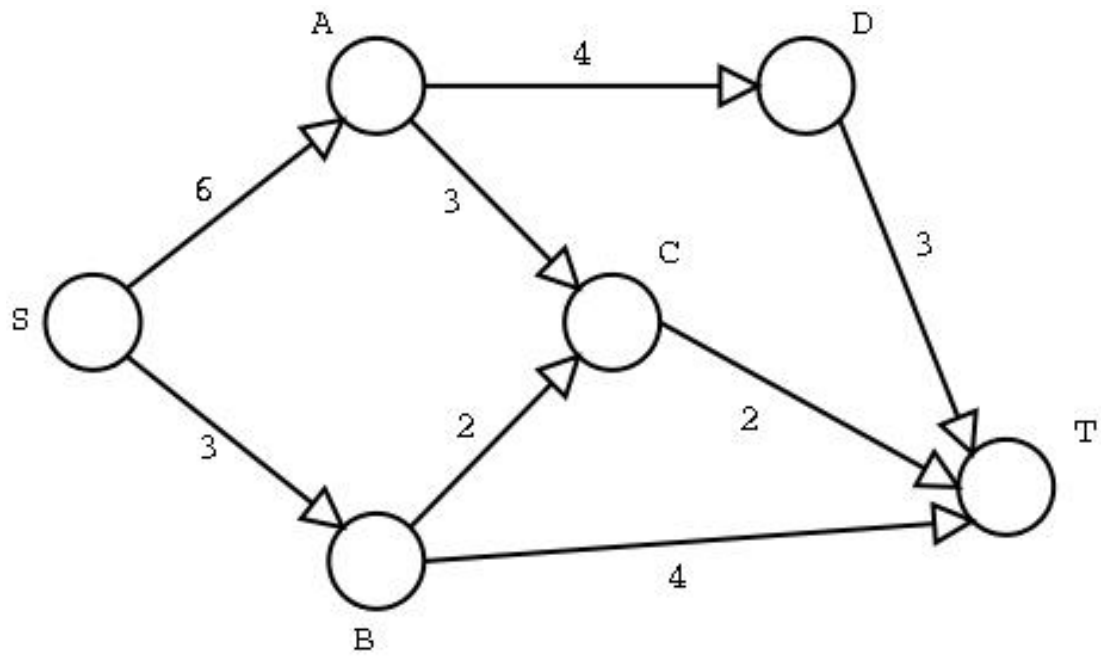
Example

- Consider a firm that has to ship haggis from Edinburgh to Southampton
- The shipping firm transports this in crates which it sends through intermediate cities
- The number of crates is limited by the size of the lorries it uses



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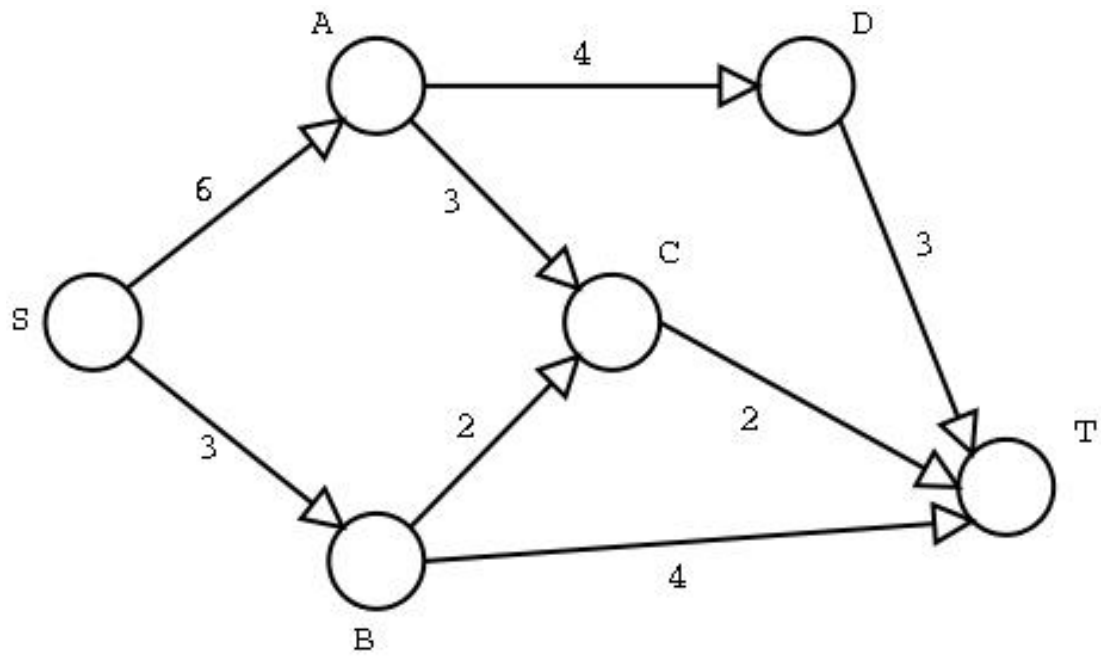

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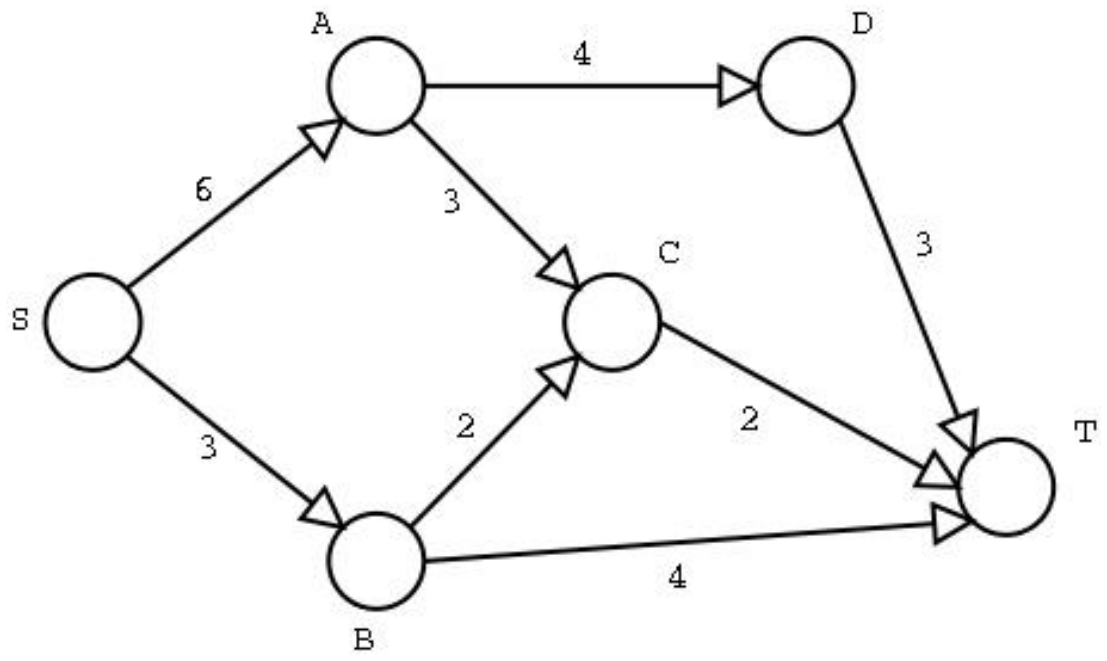

Flow

- We are given a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each edge has a capacity $c(i, j)$
- We define the flow from i to j as $f(i, j)$ with $0 \leq f(i, j) \leq c(i, j)$
- For all vertices except the source (s) and sink (t) we assume

$$\forall i \in \mathcal{V}/\{s, t\} \qquad \sum_{j \in \mathcal{V} | (i,j) \in \mathcal{E}} f(i, j) = \sum_{j \in \mathcal{V} | (i,j) \in \mathcal{E}} f(j, i)$$

  (i.e. no flow is lost from source to sink)

- We want to maximise the flow from the source

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- As set up we have a linear objective function with linear constraints
- We can therefore solve this problem with a LP-solver
- (Note the solution will typically involve a fraction flow)
- However, this is such a classic problem with a distinctive structure that we can solve it more quickly with other algorithms
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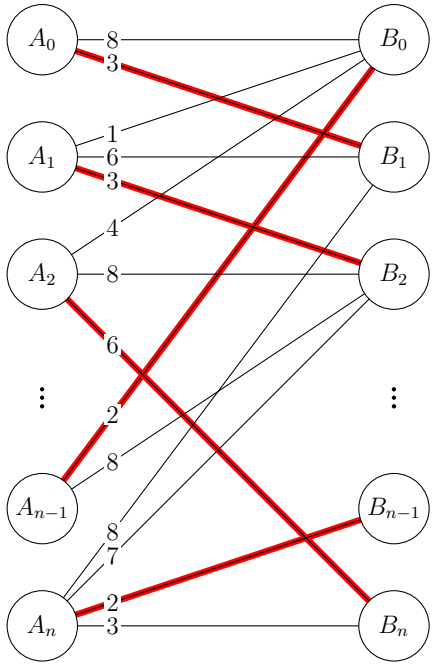
- We are given a set of n agents, $\mathcal{A}$, and n tasks, $\mathcal{T}$
- Each agent has a cost associated with performing a task $c(a, t)$
- We want to assign an agent to one task so as to minimise the total cost
- Consider a taxi firm with taxi's at 5 different locations and 5 requests to fulfil. The cost is the distance to the clients. Which taxi should go to which client?



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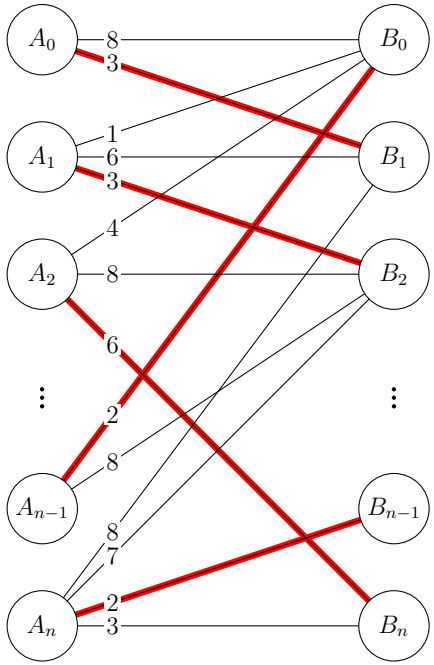

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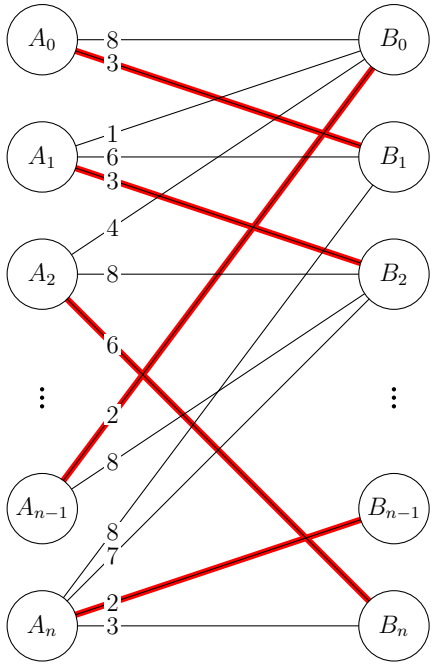

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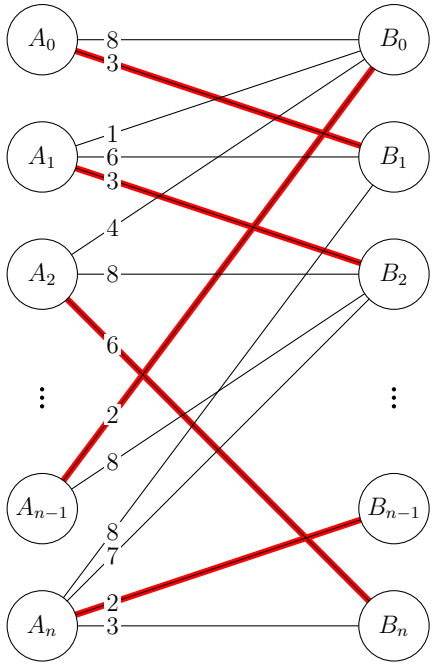

LA as LP

- The linear assignment problem can be set as a linear programming problem

$$\min_{\boldsymbol{x}} \sum_{a \in \mathcal{A}, t \in \mathcal{T}} c(a,t) x_{a,t}$$

subject to

$$\forall a \in \mathcal{A} \qquad \sum_{t \in \mathcal{T}} x_{a,t} = 1$$

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$$\forall (a,t) \in (\mathcal{A}, \mathcal{T}) \qquad x_{a,t} \geq 0$$

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